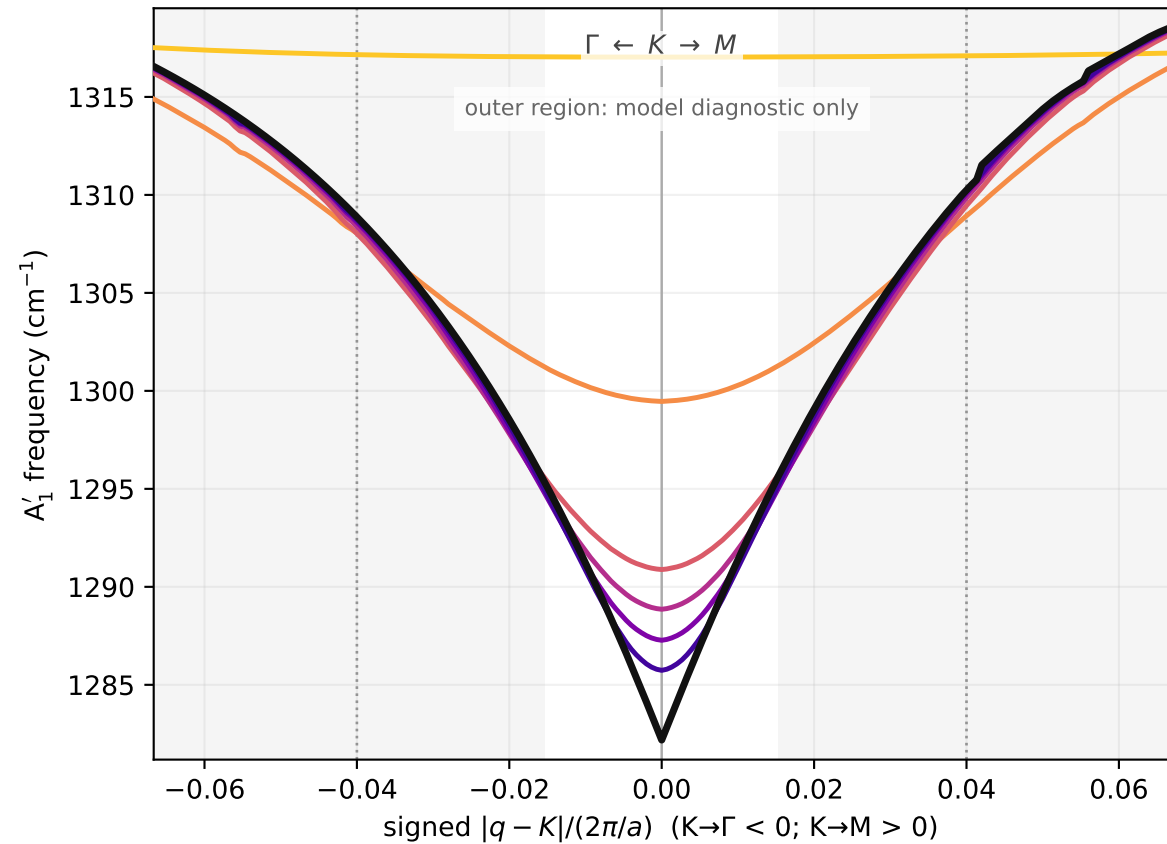
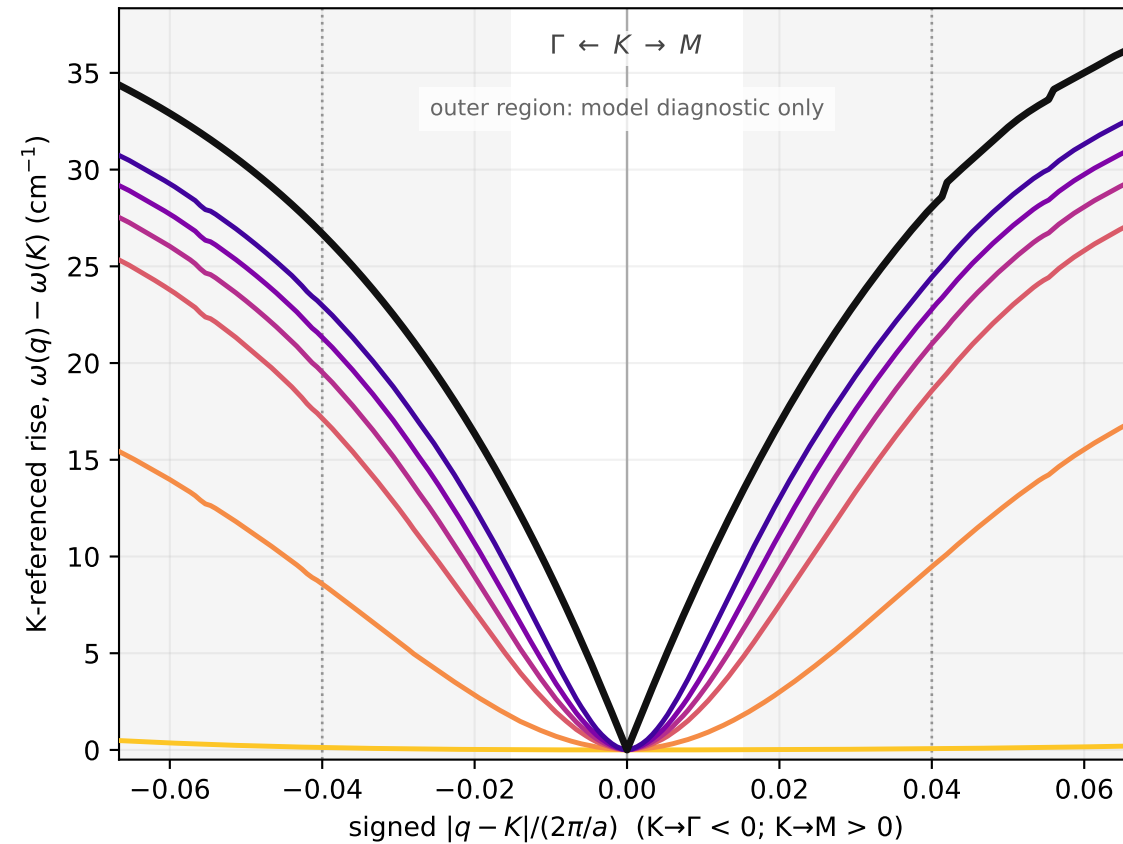


Graphene K-point A'_1 : wider and near-K electronic-smearing views

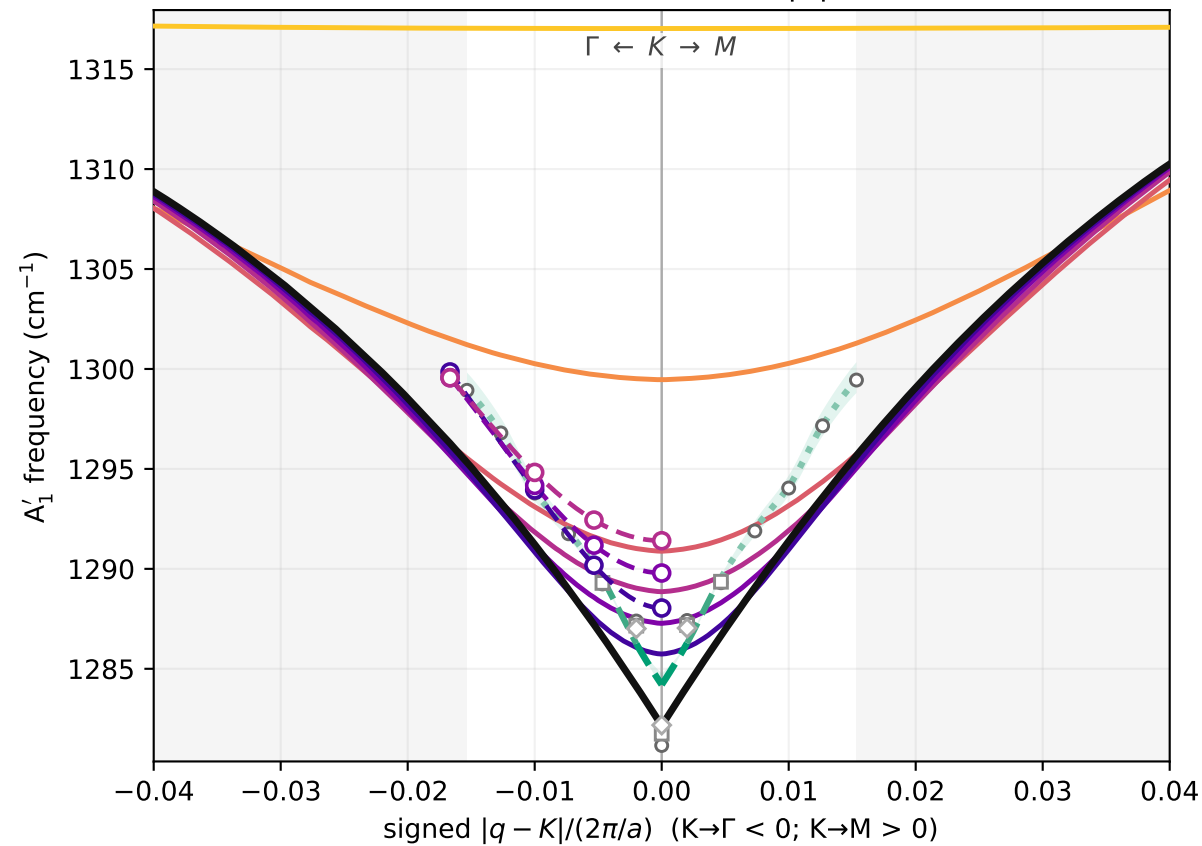
(a) Wider absolute view: $|d| \leq 0.10$



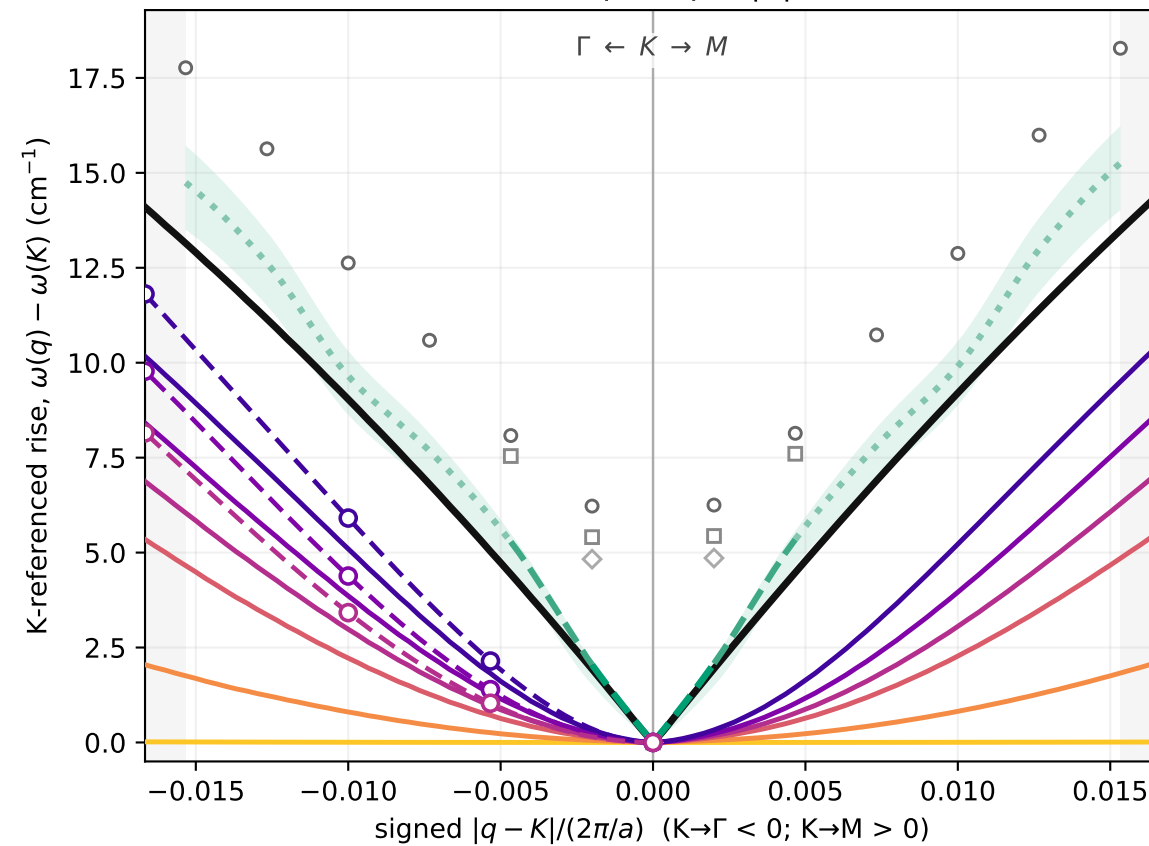
(b) Wider K-referenced view: $|d| \leq 0.10$



(c) Near-K absolute view: $|d| \leq 0.06$



(d) Near-K cusp shape: $|d| \leq 0.025$



MLIP + full EPC LR
smearing/degauss (Ry)

- 0
- 0.0019001
- 0.0028501
- 0.0038002
- 0.0050000
- 0.0100000
- 0.0200000

- zero-smearing DFPT fitted $N \rightarrow \infty$ guide
- zero-smearing guide envelope
- raw zero-smearing DFPT (k192/240/288)
- direct finite-smearing DFPT (K→Γ only)
- finite-smearing DFPT constrained fits (K→Γ only)
- outside direct zero-smearing DFPT support
- previous plotted limit: $|d| = 0.06$

Fixed lattice condition:
same static MLIP $D_{SR}(q)$ for all 7 curves;
no lattice/TDEP-temperature mixing.

Top row: direct full-EPC q evaluations to $|d|=0.10$.
Beyond $|d|=0.023$ there are no direct zero-smearing
DFPT anchors; the upper panels are model diagnostics.

A full run to $|d|=0.15$ found outer numerical wiggles;
that unstable segment is retained in data but not plotted.

Bottom row: near-K view with available DFPT evidence.
Finite-DFPT fits: $\Delta\omega = ad^2 + bd^3 + cd^4$, zero slope at K;
drawn only over measured $0 \leq d \leq 0.025$ (K→Γ).
0.020 Ry is the frozen high-smearing reference.